

FIRST® Tech Challenge

32477 Origin

Quick Start Guide

Welcome aboard, and welcome to the world of FTC!

Whether you are a newcomer who has just stepped into the robotics lab full of curiosity, or a veteran who has been through the rigors of competition and is ready to lead the team to new heights, this guide is written for you.

Building a competitive robot is never simply bolting metal and code together — it is an adventure that blends engineering logic, teamwork, and the spirit of innovation. We compiled this guide to give that adventure clear navigation.

Over past seasons we have accumulated valuable experience — and stepped on our fair share of pitfalls. We realized that a team's strength cannot rely on a few "lone heroes"; it requires unified workflows and efficient collaboration mechanisms.

The purpose of this guide is simple: refuse to reinvent the wheel.

We hope it helps newcomers skip the wandering period of confusion and quickly build a holistic understanding of FTC. At the same time, it serves as a "handbook" and "cheat sheet" for veterans — standardizing operating procedures so that every brainstorm, every part assembly, and every code

commit connects seamlessly. Only with a solid foundation can the team's growth be truly sustainable, freeing our energy for real innovation.

On the FTC field, everyone can find their own shining moment. Around the life cycle of a single robot, we have organized four closely connected tracks. Through this guide, you can explore your interests and identify your future direction:

- **3D Modeling & Design:** The bridge between imagination and reality. Before parts are physically manufactured, you will conduct precise conceptualization, assembly, and mechanical analysis in virtual space to ensure design feasibility.
- **Hardware & Build:** Turning blueprints into reality. You are the creators of the robot's skeleton and muscles — assembling gears, extrusion, and motors into a stable, powerful steel warrior on the field.
- **Programming:** Giving the robot its soul. From low-level logic to application-level algorithms, to standardized version control of team code, you are the creators who grant cold metal a smart brain.
- **Outreach & Public Relations:** The team's megaphone and diplomats. Securing sponsorship, writing engineering notebooks, promoting STEM education, and connecting with other teams. You extend this team's influence far beyond the field itself.

If you are a newcomer: do not treat this as a dry textbook — treat it as a "career roadmap." Read the whole guide first to understand what each division does, then dive deep into the area that interests you most. If you encounter unfamiliar terminology, ask a veteran anytime — this guide is your passport into the team.

If you are a veteran: use it as your "baseline." When working or mentoring newcomers, follow the standard procedures in the guide to ensure our technical knowledge is passed down accurately.

FTC is a hardcore tech competition, but it is also a story about growth and companionship. Here you will experience the frustration of mismatched

parts and the desperation of buggy code — but even more, the euphoria of solving a problem and the deep bonds forged fighting side by side with teammates.

We hope this guide becomes a reliable companion on your FTC journey. In the new season, let us uphold professionalism, experiment boldly, and verify carefully.

Enjoy the competition, enjoy creating, and let us reach new heights!

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